1. Project Overview

The Adaptive Peer Learning Platform is a web-based system designed to help university students find suitable peer learning partners based on their academic needs, skills, interests, learning preferences, and availability.
Students will create profiles containing information about the modules they are studying, topics they understand, topics they need help with, and their preferred ways and times for learning. The system will analyse this information and recommend other students who may be suitable learning partners.
The platform will use a matching algorithm to calculate a compatibility score between students and provide reasons explaining why a particular peer has been recommended.
The system will also collect feedback from

completed study interactions. This feedback will be used to improve future peer recommendations, making the matching process adaptive over time.
The system will be accessible through a web browser and will be deployed to a cloud environment so that students can access it remotely.

2. Project Objectives

The objectives of the system are to:

1. Develop a web-based platform that allows university students to connect with suitable peer learners.
2. Allow students to create and maintain profiles containing relevant academic and learning information.
3. Allow students to specify the modules they are studying and the topics or skills they

4. Allow students to identify skills and topics they can assist other students with.
5. Allow students to specify their interests, learning preferences, and availability.
6. Develop a matching system that identifies compatible peer learners.
7. Generate a compatibility score for recommended peers.
8. Provide understandable reasons explaining why students have been matched.
9. Allow students to connect with recommended peers and arrange study sessions.
10. Collect feedback from students after peer learning sessions.
11. Use previous interaction feedback to improve future peer recommendations.
12. Provide an administrative interface for managing users and relevant system information.

Deploy the completed platform to a publicly accessible cloud environment.

3. Functional Requirements

FR-01 – User Registration

The system shall allow university students to create an account using required personal and academic information.

FR-02 – User Authentication

The system shall allow registered students to securely log in and log out of the platform.

FR-03 – Student Profile Management

The system shall allow students to create, view, and update their learning profiles.
A profile shall contain relevant information such

- University
- Programme
- Year of study
- Modules
- Skills
- Learning needs
- Interests
- Learning preferences
- Availability

FR-04 – Module Management

The system shall allow students to select the modules or subjects they are currently studying.

FR-05 – Skills Management

The system shall allow students to specify skills or academic topics they are confident in and can assist other students with.

FR-06 – Learning Needs

The system shall allow students to specify topics or skills they need assistance with. Students shall be able to indicate the importance or priority of their learning needs.

FR-07 – Interests

The system shall allow students to select or enter interests that may be used as a secondary factor when determining peer compatibility.

FR-08 – Learning Preferences

The system shall allow students to specify their preferred learning methods, such as one-on-one learning, group learning, practical exercises, discussions, online learning, or in-person learning.

The system shall allow students to specify the days and times during which they are available for peer learning.

FR-10 – Peer Matching

The system shall analyse student profiles and identify potentially compatible peer learners. The matching process shall consider factors such as:

- Learning needs
- Skills
- Modules
- Availability
- Learning preferences
- Interests
- Previous interaction feedback

FR-11 – Compatibility Score

The system shall calculate a compatibility score for potential peer matches.
The score shall be represented as a percentage between 0% and 100%.

FR-12 – Match Explanation

The system shall provide reasons explaining why a particular student has been recommended.
For example:

- The student can assist with a topic the user needs help with.
- Both students are studying the same module.
- Both students have overlapping availability.
- The students have compatible learning preferences.

FR-13 – Recommended Peers

recommended peer learners to each student.
Students shall be able to view the profiles of recommended peers.

FR-14 – Connection Requests

The system shall allow a student to send a connection or study request to another student.
The receiving student shall be able to accept or decline the request.

FR-15 – Study Sessions

The system shall allow connected students to arrange study sessions.
A study session shall contain information such as:

- Participants
- Module
- Topic

- Time
- Status

FR-16 – Study Session Management

The system shall allow students to view their upcoming and previous study sessions.
Sessions shall have statuses such as:

- Pending
- Accepted
- Completed
- Cancelled

FR-17 – Session Feedback

The system shall allow students to provide feedback after completing a study session.
Feedback may include:

- Rating

- Whether the session was useful
- Whether the student would study with the peer again
- Optional comments

FR-18 – Adaptive Matching

The system shall use previous peer interaction and feedback data to influence future matching recommendations.
Positive interactions may increase compatibility between students, while consistently negative interactions may reduce future recommendations between those students.

FR-19 – Student Dashboard

The system shall provide each student with a dashboard showing relevant information, including:

- Match scores
- Connection requests
- Upcoming study sessions
- Previous sessions
- Notifications
- Profile completion

FR-20 – Notifications

The system shall notify students about relevant events, including:

- New connection requests
- Accepted or declined requests
- Study session invitations
- Upcoming study sessions
- Requests for session feedback

FR-21 – Administrator Management

The system shall provide an administrator

interface for managing the platform.
Administrators shall be able to:

- View registered students
- Manage modules
- View study sessions
- Manage inappropriate or inactive accounts
- View basic system statistics

FR-22 – Data Persistence

The system shall store user profiles, academic information, matches, connection requests, study sessions, and feedback in a persistent database.

FR-23 – Access Control

The system shall restrict access to functionality according to the user's role.
Students shall only have access to student

shall only be accessible to authorised administrators.

FR-24 – Deployment

The completed system shall be deployed to a cloud environment and made accessible through a public web URL.